STAT 527 Advanced Regression Analysis
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Diagnostics in Linear Regression

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Learning Objectives

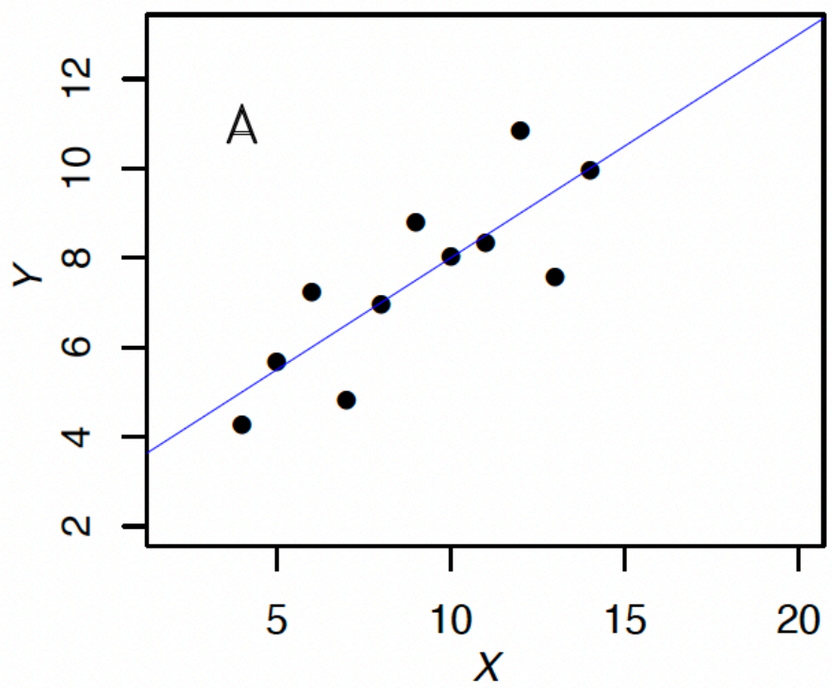
- Test for outliers using the mean shift model
- Detect influential observations using Cook's distance
- Generate diagnostic plots using R and give interpretations

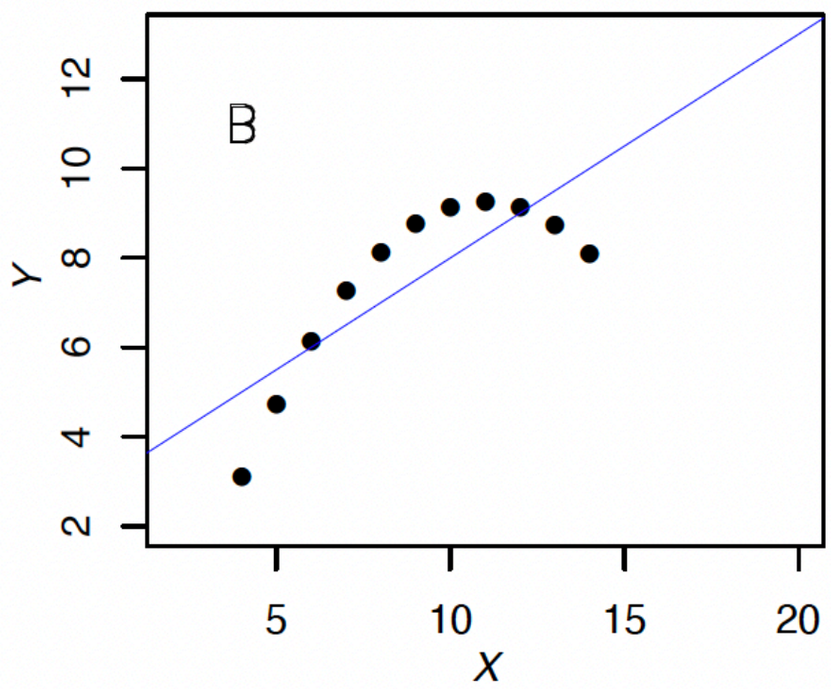
Model Diagnostics

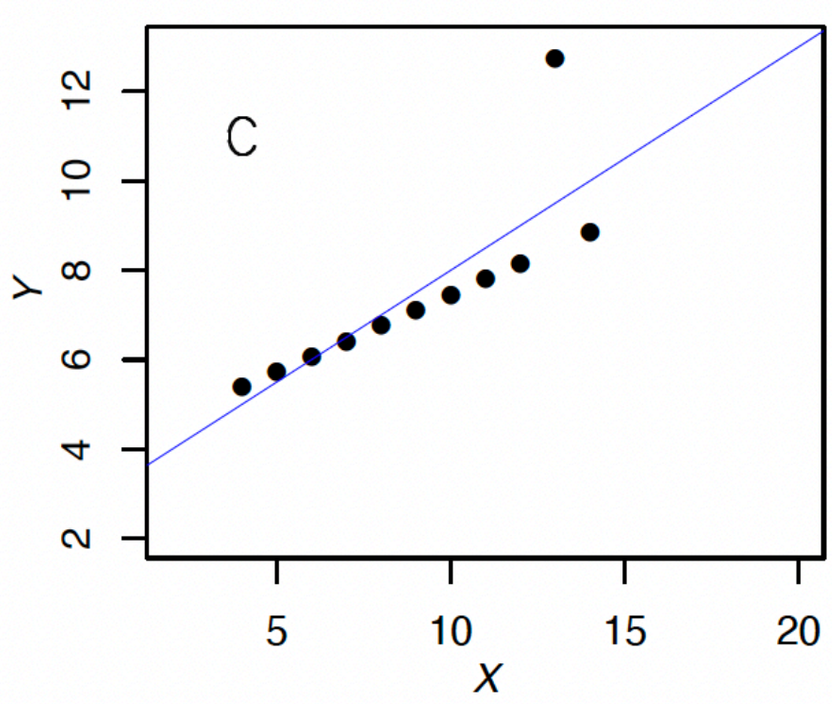
- Definition: evaluate model **assumptions** and investigate whether there are observations with a large, undue **influence** on the analysis.

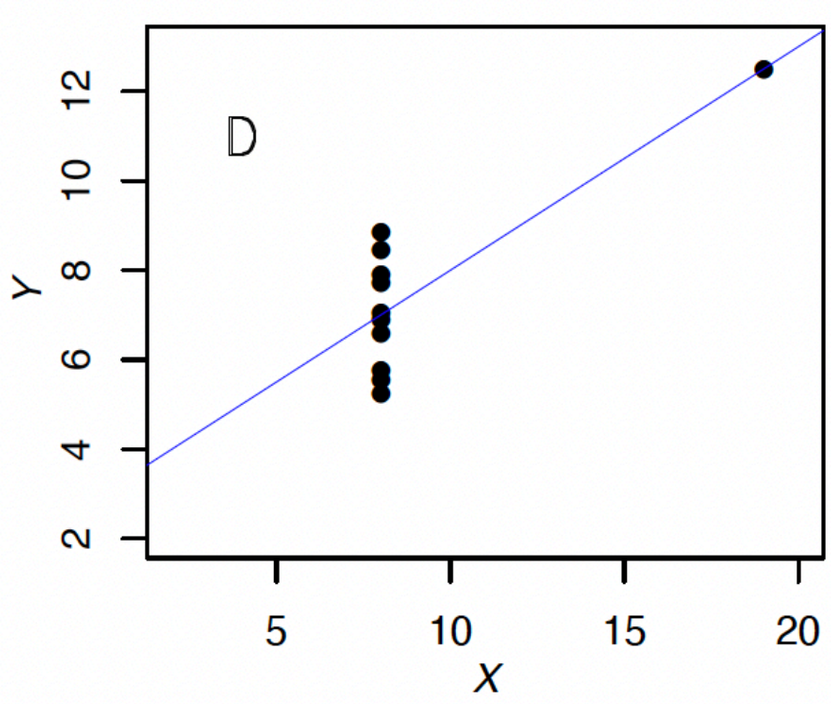
- Consider the Anscombe datasets, consisting of 4 datasets with identical sample descriptive statistics (linear regression lines), yet having different distributions and appearances.

* descriptive statistics means summarizing and describing the main features of a dataset without making conclusions about the population, in contrast to inferential statistics.


A
Y
X
2
4
6
8
10
12
5
10
15
20


B
Y
X
2
4
6
8
10
12
5
10
15
20


C
Y
X
2
4
6
8
10
12
5
10
15
20


D
Y
X
2
4
6
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12
5
10
15
20

Plots A and B

A. The simple linear regression model appears to offer a reasonably good fit. The scatterplot appears to depict a linear trend, with homoscedastic variance.

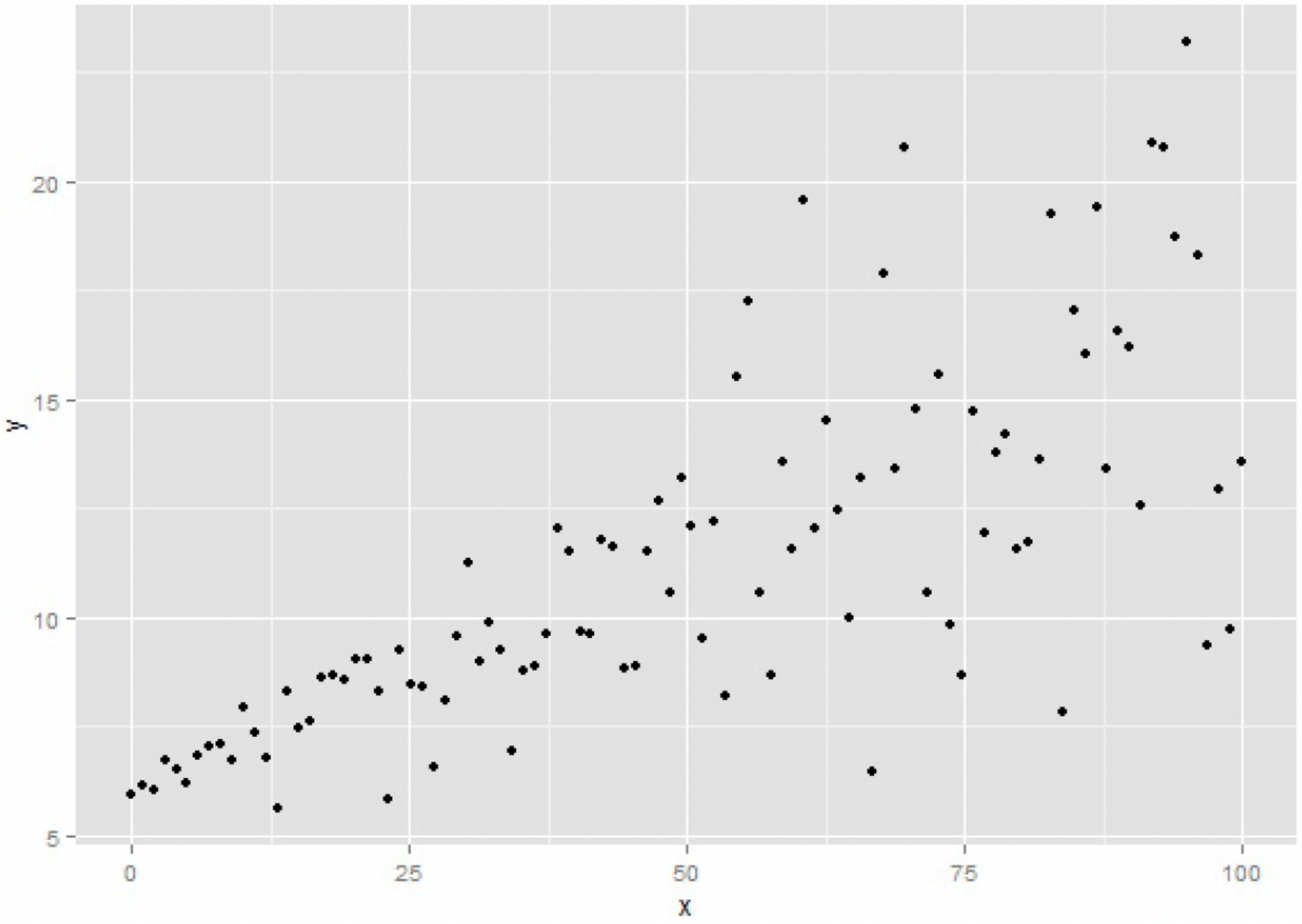
B. It is immediately seen that a linear fit is inappropriate, and a smooth curve such as a quadratic polynomial may be needed.

Plots C and D

C. A simple linear regression model seems appropriate for the data, except for a single point which has considerable influence on the fitted line. This point may be considered an outlier.

D. Data are available only for two distinct values of X, with only one observation at the highest value of X. Because of these limitations, one cannot know whether the regression of Y on X follows a line, nor if the variance is homoscedastic. Selecting a simple linear regression model in this case requires assumptions that cannot be checked. Also, the Y observation at $X = 19$, has too much influence on the fitted line.

Constant Variance?



Diagnostics Overview

Unusual Observations:

- Outliers - do not fit the model well
- Influential points - affect the fit of the model substantially

Error Assumptions:

- $E(\epsilon) = 0$
- $var(\epsilon) = \sigma^2 I$
- ϵ's are independent and normally distributed

Structure of the Model:

- Is linear model appropriate?
- Nonlinear regression?

Least Squares

Basic model in matrix notation:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where $\mathrm{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\boldsymbol{I}$.

Recall that the least squares estimate is given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

Hat (or Projection) Matrix

The fitted values $\hat{\boldsymbol{y}}$ corresponding to $\boldsymbol{y}$ are equal to

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$$

$$= \boldsymbol{H}\boldsymbol{y}$$

The matrix

$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$$

is called the **hat matrix** because it is the linear transformation of $\boldsymbol{y}$ into $\hat{\boldsymbol{y}}$.

Residuals

The residuals $\hat{\boldsymbol{\varepsilon}}$ are then given by

$$\hat{\boldsymbol{\varepsilon}} \;=\; \boldsymbol{y} - \hat{\boldsymbol{y}} \;=\; (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y}$$

Properties of $\hat{\boldsymbol{\varepsilon}}$:

We know that $\hat{\boldsymbol{\beta}}$ is an unbiased estimate of $\boldsymbol{\beta}$, which implies that $\mathrm{E}(\hat{\boldsymbol{y}}) = \boldsymbol{X}\boldsymbol{\beta} = \mathrm{E}(\boldsymbol{y})$, which implies $\mathrm{E}(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$.

Also,

$$\begin{aligned} \mathrm{var}(\hat{\boldsymbol{\varepsilon}}) \;&=\; (\boldsymbol{I} - \boldsymbol{H})\,\mathrm{var}(\boldsymbol{y})(\boldsymbol{I} - \boldsymbol{H})^T \\ &=\; \sigma^2(\boldsymbol{I} - \boldsymbol{H})(\boldsymbol{I} - \boldsymbol{H}) \;=\; \sigma^2(\boldsymbol{I} - \boldsymbol{H}) \end{aligned}$$

(using special properties of $\boldsymbol{H}$).

Leverages

Also, because the residuals are just linear combinations of the observations, they will be normally distributed if the errors are normally distributed and independent.

Note that $\mathrm{var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$.

The diagonal elements h_{ii} of $\boldsymbol{H}$ are called the **leverages**, and are given by

$$h_{ii} \;=\; \boldsymbol{x}_i^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{x}_i$$

where $\boldsymbol{x}_i^T$ is the ith row of $\boldsymbol{X}$.

They have the property that

$$\sum_{i=1}^{n} h_{ii} \;=\; rank(\boldsymbol{X}) \;=\; p'$$

High Leverage Observations

Also, for models with an intercept, the ith leverage satisfies

$$\frac{1}{n} \leq h_{ii} \leq 1. \quad \textbf{why?}$$

In addition,

$$\sum_{i=1}^{n} h_{ij} = \sum_{j=1}^{n} h_{ij} = 1$$

$$\hat{y}_i = h_{ii} y_i + \sum_{j \neq i} h_{ij} y_j.$$

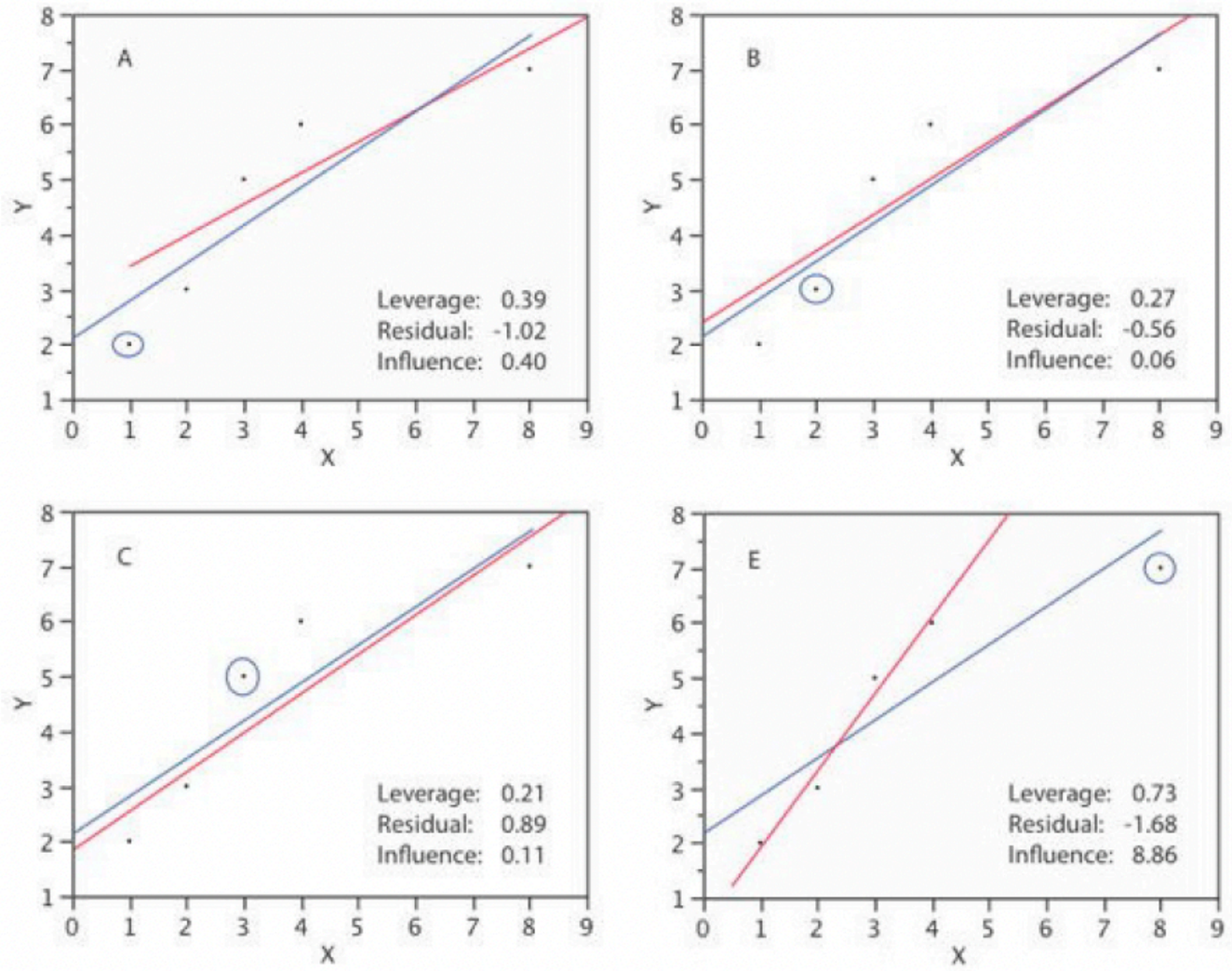
Indeed, as h_{ii} approaches 1, $\hat{y}_i$ approaches y_i.

We would then say that the ith observation has *high leverage*.

Proof of the Property

- To show $h_{ii} \leq 1$, we simply note that H is a projection (no need to assume intercept).
- To show $h_{ii} \geq \frac{1}{n}$, we do need the existence of intercept. Note that $H = \sum_{i=1}^{p'} v_i v_i^\top$, where $v_1, \ldots, v_{p'}$ are orthonormal vectors spanning $\text{col}(X)$. We can take $v_1 = \frac{1}{\sqrt{n}}(1, \ldots, 1)^\top$.

Leverage



The circled points are not included in the calculation of the red regression line. All points are included in the calculation of the blue regression line. Figure source: http://onlinestatbook.com/2/regression/influential.html

Leverage and Distance to the mean

Considering the simple linear regression case for illustration, the leverage of the ith observation is determined by the distance of x_i from $\bar{x}$ as well as the variance of x in the sample. In particular,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{SXX}$$

where

$$SXX = \sum_{j=1}^{n} (x_j - \bar{x})^2$$

In general, when $\boldsymbol{x}_i$ (the ith row of $\boldsymbol{X}$) is near the mean (that is, the independent variable values are near their means), the value of h_{ii} will be near its minimum. The farther $\boldsymbol{x}_i$ is from the mean, the larger h_{ii} will tend to be.

Studentized Residuals

- We have seen that $\mathrm{var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$ will be small when h_{ii} is large
- Cases in which $\boldsymbol{x}_i$ (the ith row of $\boldsymbol{X}$) is near the mean will tend to have large variance and cases far from the mean will tend to have small variance.
- However, this may be undesirable if the linear regression model becomes inadequate far from the mean, because the estimator puts greater emphasis on fitting these values more precisely, even though the model is inadequate there.

Studentized Residuals

We consider a standardized version of the residuals, called **internally Studentized residuals**, obtained by dividing the $\hat{\varepsilon}_i$ by their standard errors:

$$r_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$$

(In R, these are called *standardized residuals*.)

When the model is correct, $\mathrm{E}(r_i) = 0$ and $\mathrm{var}(r_i) \approx 1$.

why?

Property the Studentized Residuals

- Unlike the case of t-statistic, we do not have independence of $\hat{\epsilon}_i$ and $\hat{\sigma}^2$.
- However, when the noise has a symmetric distribution, we see that $\frac{\hat{\epsilon}_i}{\hat{\sigma}}$ also has a symmetric distribution, so $\mathbb{E}[r_i] = 0$.

Savings Data Revisited

Let's take a look at these ideas using the savings data.

```
> library(faraway)

> # Fit full model for savings data, and examine residuals and predicted vals

> mod <- lm(sr ~ pop15 + pop75 + dpi + ddpi, data=savings)

> plot(fitted(mod), residuals(mod))
```

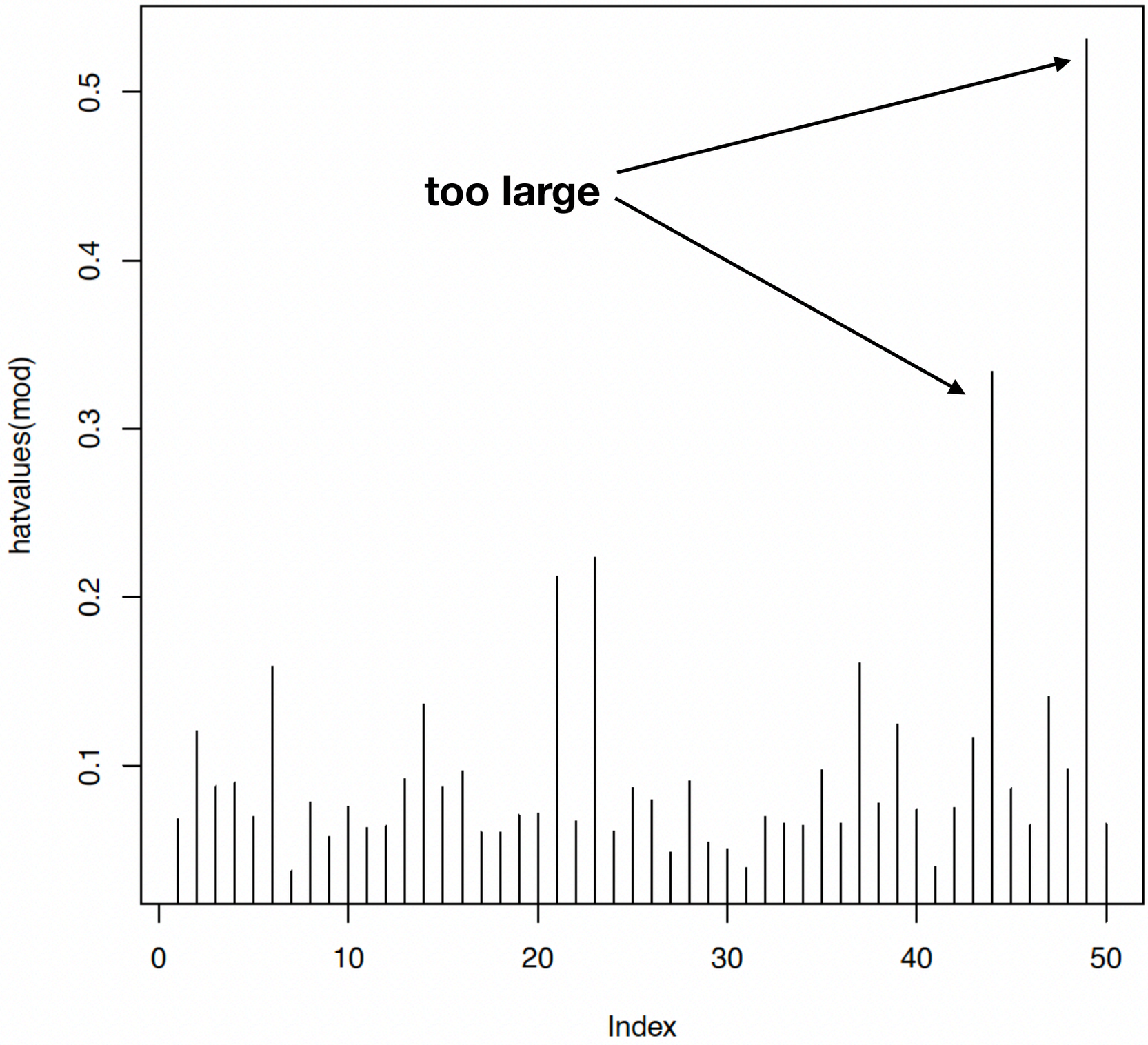
Leverages

```
> # Examine leverages

> hatvalues(mod)
    Australia       Austria       Belgium       Bolivia        Brazil
   0.06771343    0.12038393    0.08748248    0.08947114    0.06955944
       Canada         Chile         China      Colombia    Costa Rica
   0.15840239    0.03729796    0.07795899    0.05730171    0.07546780
      Denmark       Ecuador       Finland        France       Germany
   0.06271782    0.06372651    0.09204246    0.13620478    0.08735739
       Greece     Guatamala      Honduras       Iceland         India
   0.09662073    0.06049212    0.06008079    0.07049590    0.07145213
      Ireland         Italy         Japan         Korea    Luxembourg
   0.21223634    0.06651170    0.22330989    0.06079915    0.08634787
        Malta        Norway   Netherlands   New Zealand     Nicaragua
   0.07940290    0.04793213    0.09061400    0.05421789    0.05035056
        Panama      Paraguay          Peru   Philippines      Portugal
   0.03897459    0.06937188    0.06504891    0.06425415    0.09714946
   South Africa South Rhodesia         Spain        Sweden   Switzerland
   0.06510405    0.16080923    0.07732854    0.12398898    0.07359423
        Turkey       Tunisia United Kingdom  United States     Venezuela
   0.03964224    0.07456729    0.11651375    0.33368800    0.08628365
        Zambia       Jamaica       Uruguay         Libya      Malaysia
   0.06433163    0.14076016    0.09794717    0.53145676    0.06523300

> plot(hatvalues(mod), type="h")  # vertical lines to each leverage value
```

generate histogram instead of points


too large
hatvalues(mod)
0.1
0.2
0.3
0.4
0.5
0
10
20
30
40
50
Index

How Large is too Large?

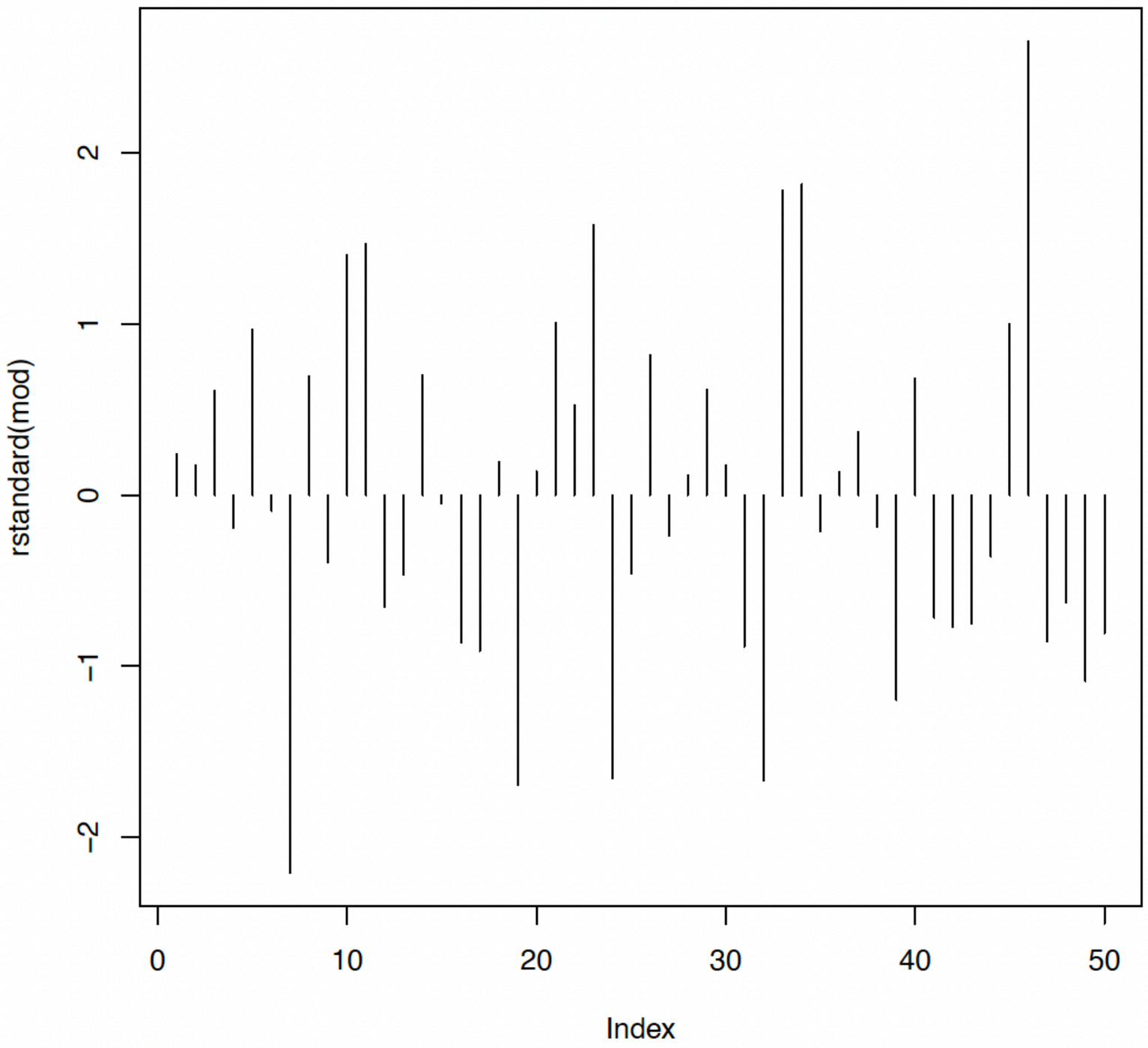
- Recall that $\frac{1}{n} \leq h_{ii} \leq 1$.
- Since $\sum_{i=1}^{n} h_{ii} = p'$, a rough rule is that $h_{ii} \geq \frac{2p'}{n}$ should be looked at more closely, although "this tends to draw attention to too many points" (Faraway, "Linear Models in R")
- What to do: decide whether to keep the sample or keep the linear model assumption.

Studentized Residuals

```
> # Examine internally Studentized residuals

> rstandard(mod)
     Australia        Austria        Belgium        Bolivia         Brazil
    0.23520105     0.17282943     0.61085760    -0.19245030     0.96858807
        Canada          Chile          China       Colombia     Costa Rica
   -0.09083873    -2.20907436     0.69453131    -0.39319153     1.40168682
       Denmark        Ecuador        Finland         France        Germany
    1.46686216    -0.65379142    -0.46394723     0.70042898    -0.04974135
        Greece       Guatamala       Honduras        Iceland          India
   -0.86217889    -0.91031261     0.19259259    -1.69401854     0.13881900
       Ireland          Italy          Japan          Korea      Luxembourg
    1.00475012     0.52442520     1.57595468    -1.65713877    -0.45967116
         Malta         Norway    Netherlands    New Zealand      Nicaragua
    0.81536209    -0.23495632     0.11735008     0.61802723     0.17443311
        Panama       Paraguay           Peru    Philippines       Portugal
   -0.88366877    -1.66987256     1.77851567     1.81461452    -0.21267488
   South Africa South Rhodesia          Spain         Sweden    Switzerland
    0.13140922     0.37072635    -0.18374340    -1.19700295     0.67944806
         Turkey        Tunisia United Kingdom  United States      Venezuela
   -0.71532499    -0.77031393    -0.75327449    -0.35811077     0.99934066
         Zambia        Jamaica        Uruguay          Libya       Malaysia
    2.65091534    -0.85634746    -0.62681420    -1.08705199    -0.80805950

> plot(rstandard(mod), type="h")
```



If assumptions are correct, the studentized residuals have approximately constant variance for all x.

Outliers

In regression analysis, the model assumptions are supposed to apply to all of the observations. Sometimes one or more observations to have observed responses that do not appear to satisfy the same model as the bulk of the data. These observations are referred to as **outliers**.

[Illustrate ...]

In regression, an observation is judged to be an outlier on the basis of its (standardized) residual. Outliers have residuals that are far from 0, relative to the other observations.

The Mean Shift Test

We develop a statistical test for a particular way in which an outlier can occur.

Suppose the ith observation is a candidate for an outlier.

We assume that the model

$$y_j = \boldsymbol{x}_j^T \boldsymbol{\beta} + \varepsilon_j$$

holds for $j \neq i$, and for the ith observation the model is

$$y_i = \boldsymbol{x}_i^T \boldsymbol{\beta} + \delta + \varepsilon_i$$

If $\delta \neq 0$, it shifts the mean of y_i and can create an outlier.

Mean Shift Model

We can test the null hypotheses $\delta = 0$, and if we reject this hypothesis, the ith observation will be considered an outlier.

This *mean shift model* for the ith observation can be written for all of the observations as follows.

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{U}\delta + \boldsymbol{\varepsilon}$$

where $\boldsymbol{U}$ is a vector for which all components are 0 except for the ith component which is 1.

Mean Shift t-statistic

One can readily see how to compute $\hat{\delta}$ and $se(\hat{\delta})$ by noticing that we can rewrite the mean-shift model as follows:

$$\boldsymbol{y} = \boldsymbol{X}_*\boldsymbol{\beta}_* + \boldsymbol{\varepsilon}$$

where $\boldsymbol{X}_* = [\boldsymbol{X}, \boldsymbol{U}]$ and $\boldsymbol{\beta}_* = (\boldsymbol{\beta}^T, \delta)^T$.

Then $\hat{\delta}$ is obtained from the last component of

$$\hat{\boldsymbol{\beta}}_* = (\boldsymbol{X}_*^T\boldsymbol{X}_*)^{-1}\boldsymbol{X}_*^T\boldsymbol{y}$$

The standard error of $\hat{\delta}$ is obtained from

$$\widehat{\mathrm{var}}(\hat{\boldsymbol{\beta}}_*) = \hat{\sigma}_*^2(\boldsymbol{X}_*^T\boldsymbol{X}_*)^{-1}$$

RSS/(n-p'-1)

Mean Shift t-statistic

To test the null hypothesis that $\delta = 0$, we find the least squares estimate $\hat{\delta}$ and compute

$$t = \hat{\delta}/se(\hat{\delta})$$

and find the p-value using a t-distribution with $n - p' - 1$ degrees of freedom as the reference distribution.

$t-$statistic and externally Studentized residual

can be calculated in one pass

An interesting fact is that the t-statistic is equal to the *externally Studentized residual*

$$t_i = r_i \left(\frac{n-p'-1}{n-p'-r_i^2} \right)^{1/2} = \frac{\hat{\varepsilon}_i}{\hat{\sigma}_{(i)}\sqrt{1-h_{ii}}}$$

internally studentized residual

Here $\hat{\sigma}_{(i)}$ refers to the estimate of σ obtained when the ith observation is left out.

Testing for Outliers

test is a priori, but after analyzing data

- We would want to perform this outlier test for all observations that are candidate outliers, one at a time.
- However, simply performing the test on the observations with largest observed residuals would be an example of *data snooping*, unless somehow these observations were identified before data collection.
- To be certain that the overall Type I error rate is no greater than α, *Bonferroni's Inequality* may be used.

  If each of the n observations is viewed as a candidate outlier (to be conservative), it would be tested individually at level α/n.

Back to the savings data set

```
> # Examine mean-shift t-statistics (externally Studentized residuals)

> rstudent(mod)
     Australia        Austria        Belgium       Bolivia         Brazil
    0.23271611     0.17095506     0.60655220    -0.19037831     0.96790816
         Canada          Chile          China      Colombia    Costa Rica
   -0.08983197    -2.31342946     0.69048169    -0.38946778     1.41731062
        Denmark        Ecuador        Finland        France       Germany
    1.48644473    -0.64957871    -0.45986445     0.69640933    -0.04918692
         Greece      Guatamala       Honduras       Iceland         India
   -0.85967533    -0.90854545     0.19051919    -1.73119989     0.13729730
        Ireland          Italy          Japan          Korea    Luxembourg
    1.00485886     0.52015744     1.60321582    -1.69103214    -0.45560591
         Malta         Norway    Netherlands    New Zealand      Nicaragua
    0.81227407    -0.23247367     0.11605663     0.61373189     0.17254242
         Panama       Paraguay           Peru    Philippines       Portugal
   -0.88147653    -1.70488128     1.82391409     1.86382587    -0.21040432
    South Africa South Rhodesia          Spain         Sweden    Switzerland
    0.12996586     0.36714512    -0.18175853    -1.20293404     0.67532922
          Turkey        Tunisia United Kingdom  United States      Venezuela
   -0.71138840    -0.76677907    -0.74959873    -0.35461507     0.99932569
          Zambia        Jamaica        Uruguay          Libya       Malaysia
    2.85355834    -0.85376418    -0.62253411    -1.08930326    -0.80489153
```

savings data set

```
> # Bonferroni-adjusted critical value is qt(alpha/2n, df=n-p'-1, lower=FALSE)

> critval <- qt(0.05/(2*nobs(mod)), df=df.residual(mod)-1, lower=FALSE)

> critval
[1] 3.525801

> which(abs(rstudent(mod)) > critval)
named integer(0)

>   # so none exceed the Bonferroni critical value
```

Influence of Individual Observations

If the fit of a model depends too much on any one observation, the fit must not be very stable.

Observations whose removal greatly affects the analysis are called **influential**.

One straightforward method for measuring influence is to fit the model with and without the ith observation, and observe the difference.

We use the subscript "(i)" to denote objects which exclude the ith observation. For instance, the least squares estimate of $\boldsymbol{\beta}$, excluding the ith observation, is

$$\hat{\boldsymbol{\beta}}_{(i)} = \left(\boldsymbol{X}_{(i)}^T \boldsymbol{X}_{(i)}\right)^{-1} \boldsymbol{X}_{(i)}^T \boldsymbol{y}_{(i)}$$

Cook's Distance

statistician Dennis Cook

Cook's Distance D_i can be used to measure the influence of the ith observation:

$$D_i = \frac{(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})^T (\boldsymbol{X}^T \boldsymbol{X})(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})}{p' \hat{\sigma}^2}$$

Defining $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{y}}_{(i)} = \boldsymbol{X}\hat{\boldsymbol{\beta}}_{(i)}$ gives

$$D_i = \frac{(\hat{\boldsymbol{y}}_{(i)} - \hat{\boldsymbol{y}})^T (\hat{\boldsymbol{y}}_{(i)} - \hat{\boldsymbol{y}})}{p' \hat{\sigma}^2}$$

Cook's Distance

To compute D_i without refitting the model n times, we use the result that

$$D_i = \frac{1}{p'} r_i^2 \left(\frac{h_{ii}}{1 - h_{ii}} \right).$$

proof of equivalence: p161 Sen, Srivastava

The magnitude of D_i can be interpreted in terms of confidence ellipsoids. For instance, by comparison with an F-distribution (in which the median is about 1), a value $D_i = 1$ would move $\hat{\boldsymbol{\beta}}_{(i)}$ to about the edge of a 50% confidence ellipsoid for $\boldsymbol{\beta}$, as computed from the complete data.

A *rule-of-thumb* is to consider observations with $D_i \geq 1$ as highly influential.

Savings Data Set

Let's look at Cook's Distances in the savings data set ...

```
> # Examine Cook's distances

> cooks.distance(mod)
     Australia        Austria        Belgium        Bolivia         Brazil
  8.035888e-04   8.175997e-04   7.154674e-03   7.278744e-04   1.402735e-02
        Canada          Chile          China       Colombia     Costa Rica
  3.106199e-04   3.781324e-02   8.156984e-03   1.879460e-03   3.207537e-02
       Denmark        Ecuador        Finland         France        Germany
  2.879580e-02   5.818699e-03   4.364051e-03   1.547176e-02   4.736572e-05
        Greece       Guatamala       Honduras        Iceland          India
  1.590102e-02   1.067111e-02   4.741920e-04   4.352902e-02   2.965778e-04
       Ireland          Italy          Japan          Korea     Luxembourg
  5.439637e-02   3.919100e-03   1.428162e-01   3.555386e-02   3.993882e-03
         Malta         Norway    Netherlands    New Zealand      Nicaragua
  1.146827e-02   5.558570e-04   2.744377e-04   4.379219e-03   3.226479e-04
        Panama       Paraguay           Peru    Philippines       Portugal
  6.333674e-03   4.157229e-02   4.401457e-02   4.522120e-02   9.733900e-04
  South Africa South Rhodesia          Spain         Sweden    Switzerland
  2.405063e-04   5.267290e-03   5.659085e-04   4.055963e-02   7.334746e-03
        Turkey        Tunisia United Kingdom  United States      Venezuela
  4.224370e-03   9.562447e-03   1.496628e-02   1.284481e-02   1.886141e-02
        Zambia        Jamaica        Uruguay          Libya       Malaysia
  9.663275e-02   2.402677e-02   8.532329e-03   2.680704e-01   9.113404e-03
```

```
> which(cooks.distance(mod) >= 1)
named integer(0)

> plot(cooks.distance(mod), type="h")
```

Leverage vs Cook's Distance

The *leverage* h_{ii} measures the *potential influence* of observation i.

Cook's distance D_i measures the *actual influence* of observation i.

Summary of R Diagnostic Functions

leverages (h_{ii}):	`hatvalues`
internally Studentized residuals (r_i):	`rstandard`
externally Studentized residuals (t_i):	`rstudent`
Cook's distance (D_i):	`cooks.distance`

Each takes the `lm`-fitted model as its argument. Use the `help` function in R to learn more.

Diagnostic Plots

Plots of the residuals (with other diagnostic quantities) are the main tool for diagnosing violations of the assumptions.

R provides a convenient set of plots, by default.

Consider again the savings data ...

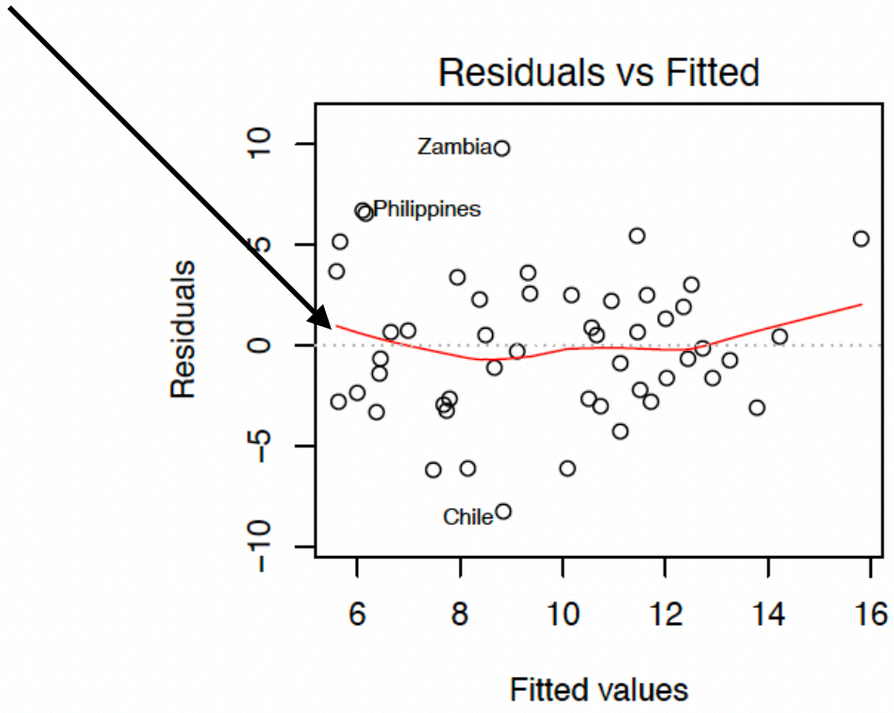
```
> # Obtain diagnostic plots

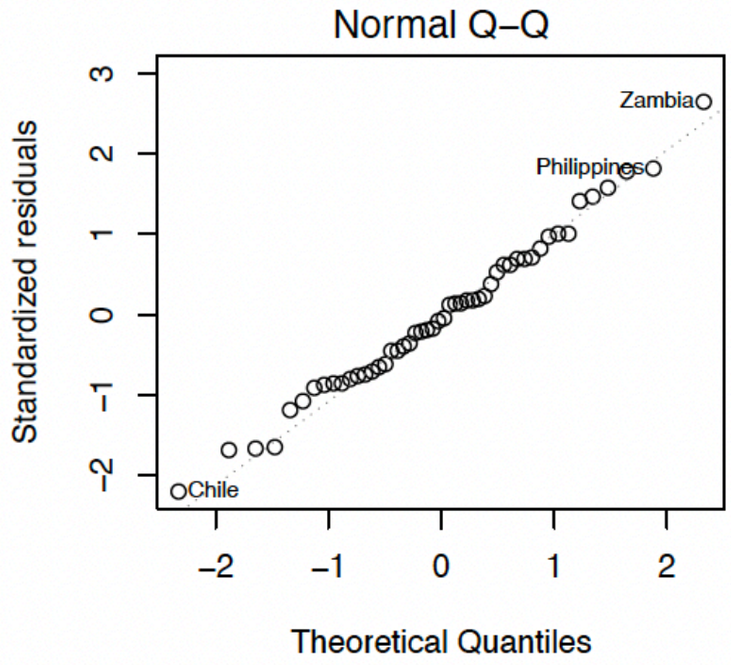
> par(mfrow=c(2,2))  # sets up a 2-by-2 plot grid for ...
> plot(mod)  # ... the 4 diagnostic plots produced by default
```

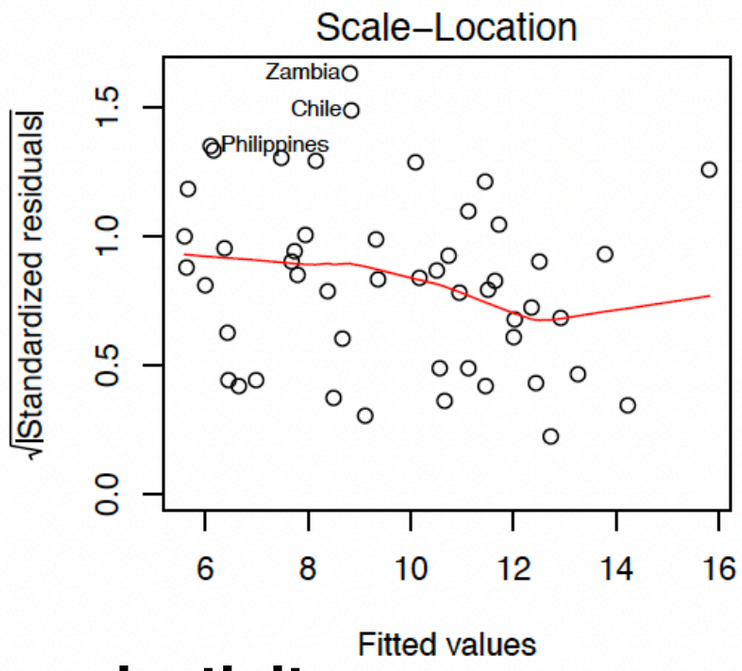
plot(lm(y~x))

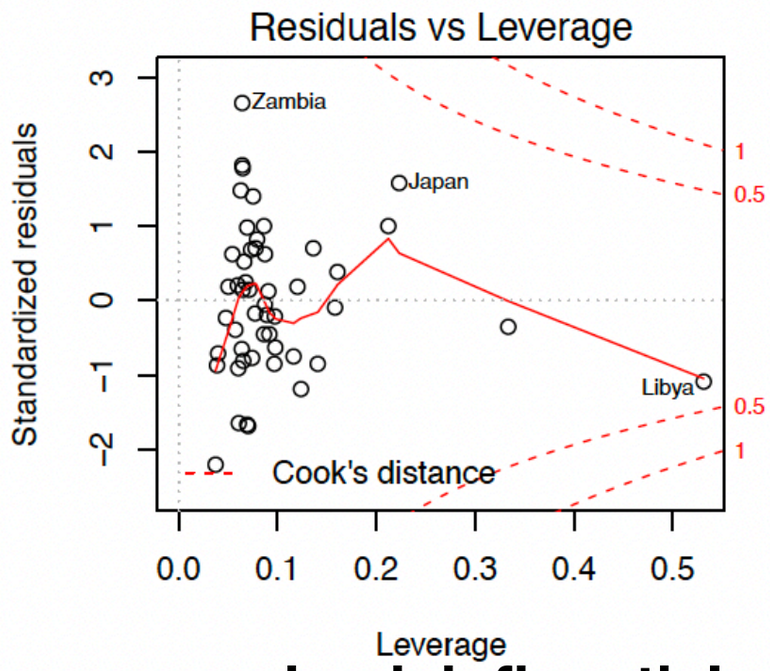
trend: nonlinear or flat?

normality affects CI, but not much on \hat{\beta} or MSE









check homoscedesticity

check influential points and outliers

Diagnostic Plots - Summary

- The **residuals-versus-fitted** plot checks the linearity and constant variance (homoscedasticity) assumptions. Trend should be roughly flat with equal vertical spread when there are no problems.

[Illustrate ...]

Diagnostic Plots - Summary

- The **normal Q-Q** plot checks normality of the distribution of the errors. The points should approximate a straight line when there is no problem with normality.
- The **scale-location** plot checks the assumption of constant variance (homoscedasticity). Trend should be roughly flat when there are no problems with variance.
- The **residuals versus leverage** plot helps to check both leverage (directly) and influence (with the contour lines for Cook's distance).

Each plot uses residuals, so it can also be used to find outliers.

Assessment of Normality

A standard graphical method for inspecting the assumption of normality makes use of **probability (Q-Q) plots**. For a sample z_1, z_2,...,z_n, we wish to assess normality.

1. Order the zs to obtain the sample order statistics

$$z_{(1)} \leq z_{(2)} \leq \cdots \leq z_{(n)}$$

2. Let $u_{(i)} = (\mathrm{E}(z_{(i)}) - \mu)/\sigma$, under the assumption that the zs are normal. A good approximation is

$$u_{(i)} = \Phi^{-1}\left(\frac{i - 3/8}{n + 1/4}\right)$$

where Φ is the cumulative distribution function of a standard normal distribution.

3. If the z's are normal, plotting the $z_{(i)}$s versus the $u_{(i)}$s should approximate a straight line. Substantial departure from this indicates non-normality (or outliers).

Additive Smoothing

- the 3/8 and 1/4 constants in the slide come from additive smoothing, which accounts for the prior belief of the distribution, see e.g.

https://en.wikipedia.org/wiki/Additive_smoothing

How to Produce Q-Q plots

```
data <- rnorm(100)  # generate 100 random values from a standard normal distribution

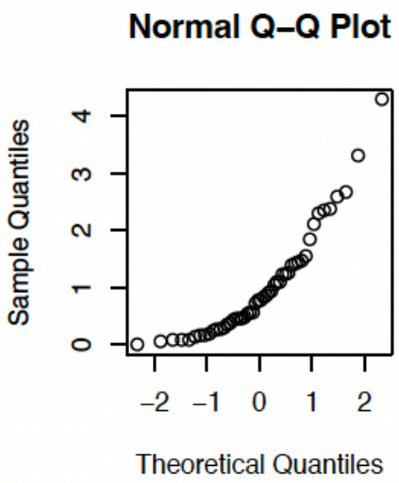
qqnorm(data)

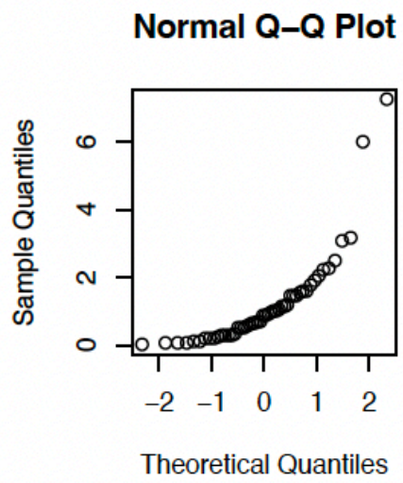
qqline(data)  # This adds a reference line which can help in determining how close the data is to normality.
```

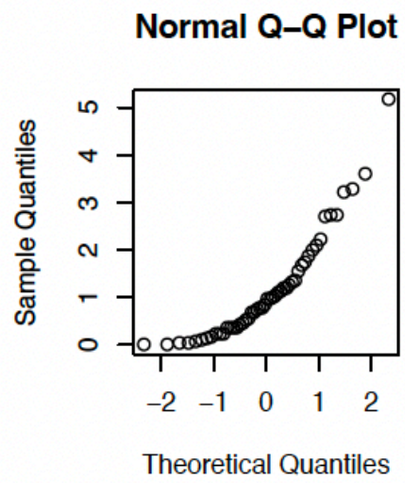
Let's try some normal Q-Q plots for sample of $n = 50$ from various distributions. We'll try 9 data sets from each to get a sense of the variability in the appearance of these plots.

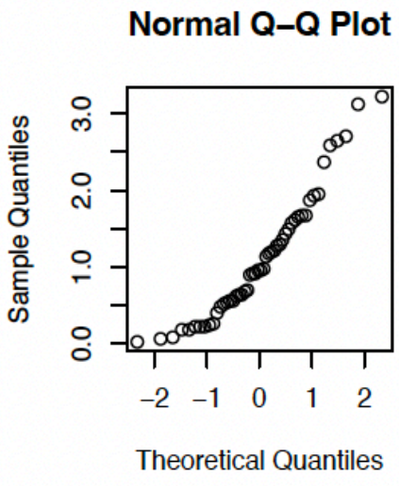
```
### first try exponential dist (lambda=1)

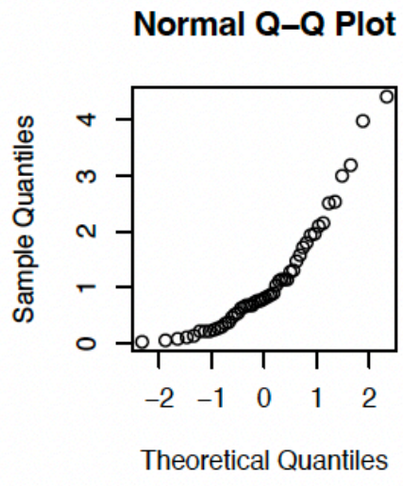
> par(pty="s")              # set up for SQUARE plots
> par(mfrow=c(3,3))         # in a 3 x 3 grid
> for(i in 1:9){
>   qqnorm(rexp(50))        # form Q-Q plot for sim. data
> }
```

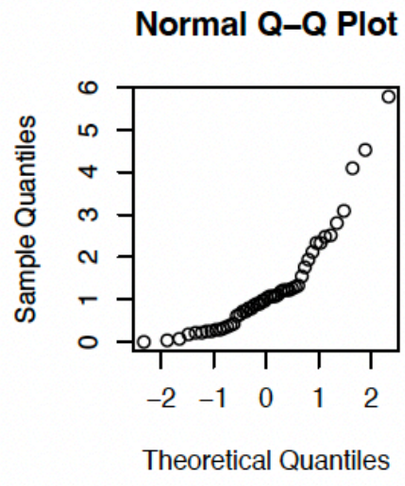

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 1 2 3 4

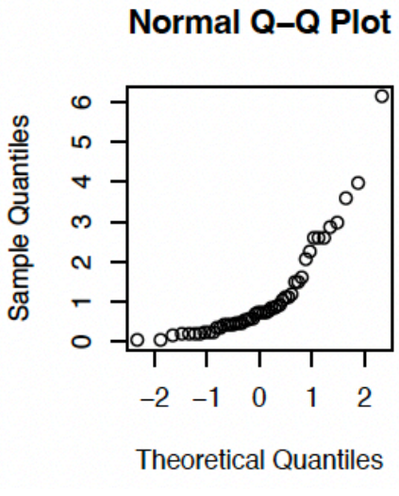

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 2 4 6

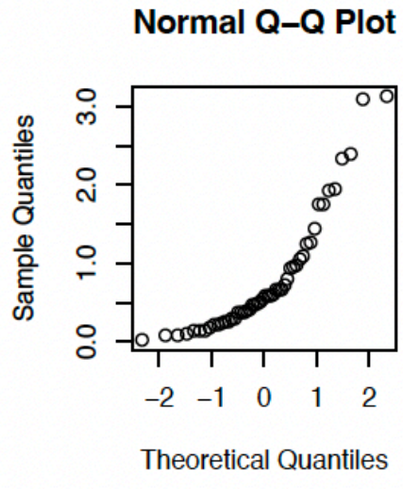

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 1 2 3 4 5

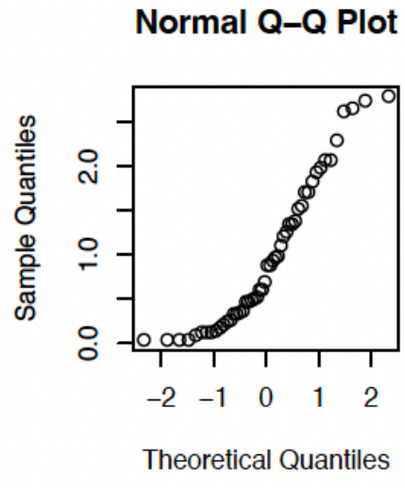

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0.0 1.0 2.0 3.0


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 1 2 3 4


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 1 2 3 4 5 6

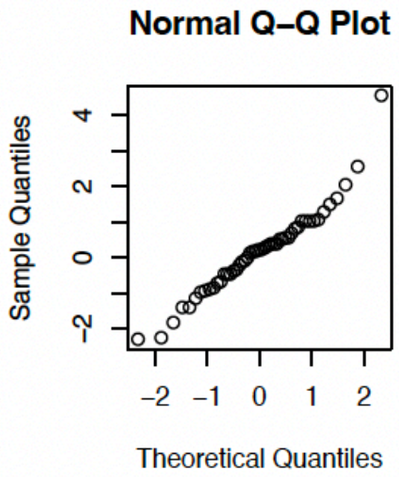

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0 1 2 3 4 5 6

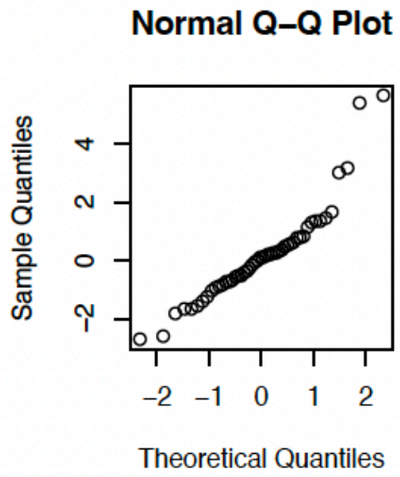

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0.0 1.0 2.0 3.0

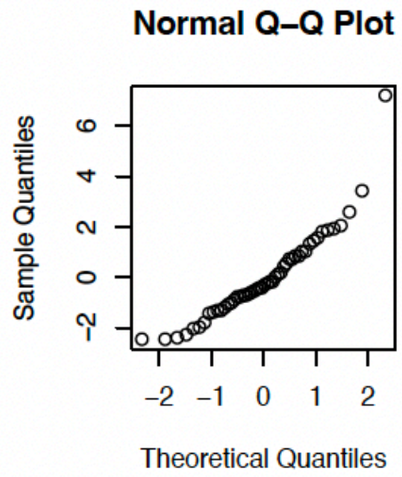

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0.0 1.0 2.0

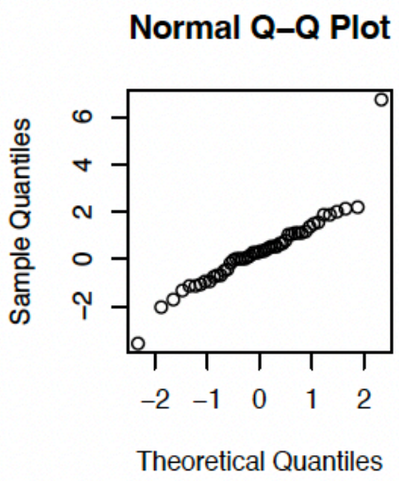
```
## now try t-dist with 3 df

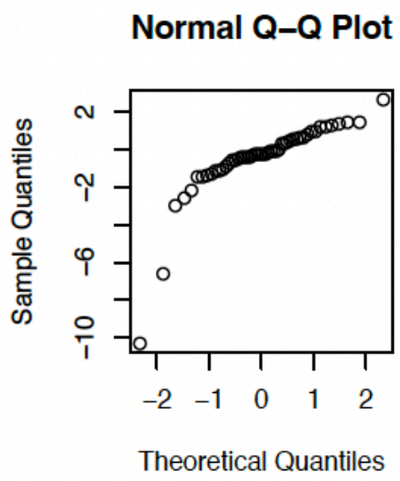
> for(i in 1:9){
>   qqnorm(rt(50,df=3))
> }
```

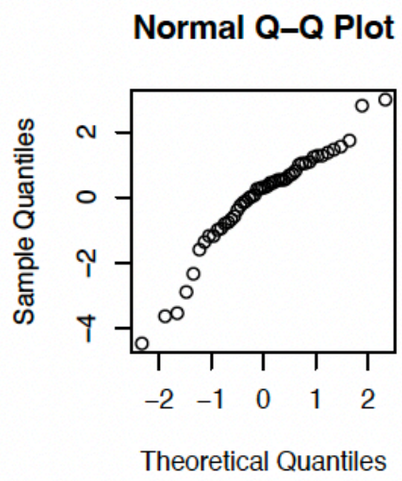

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles

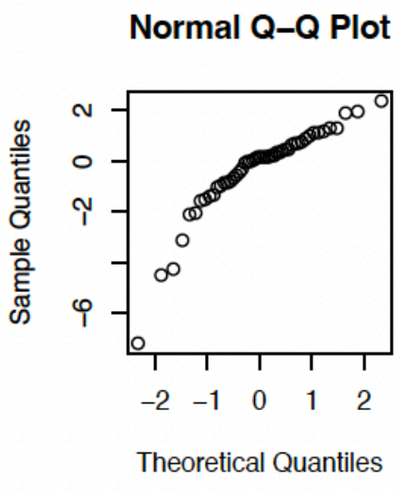

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles

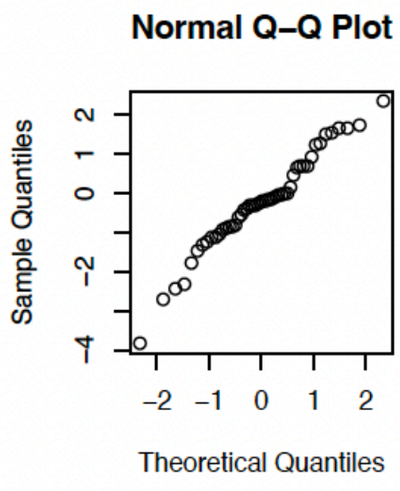

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles

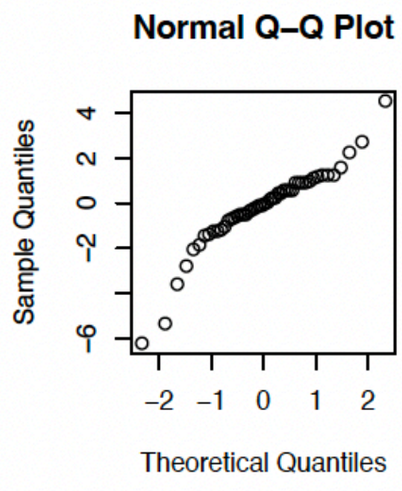

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles

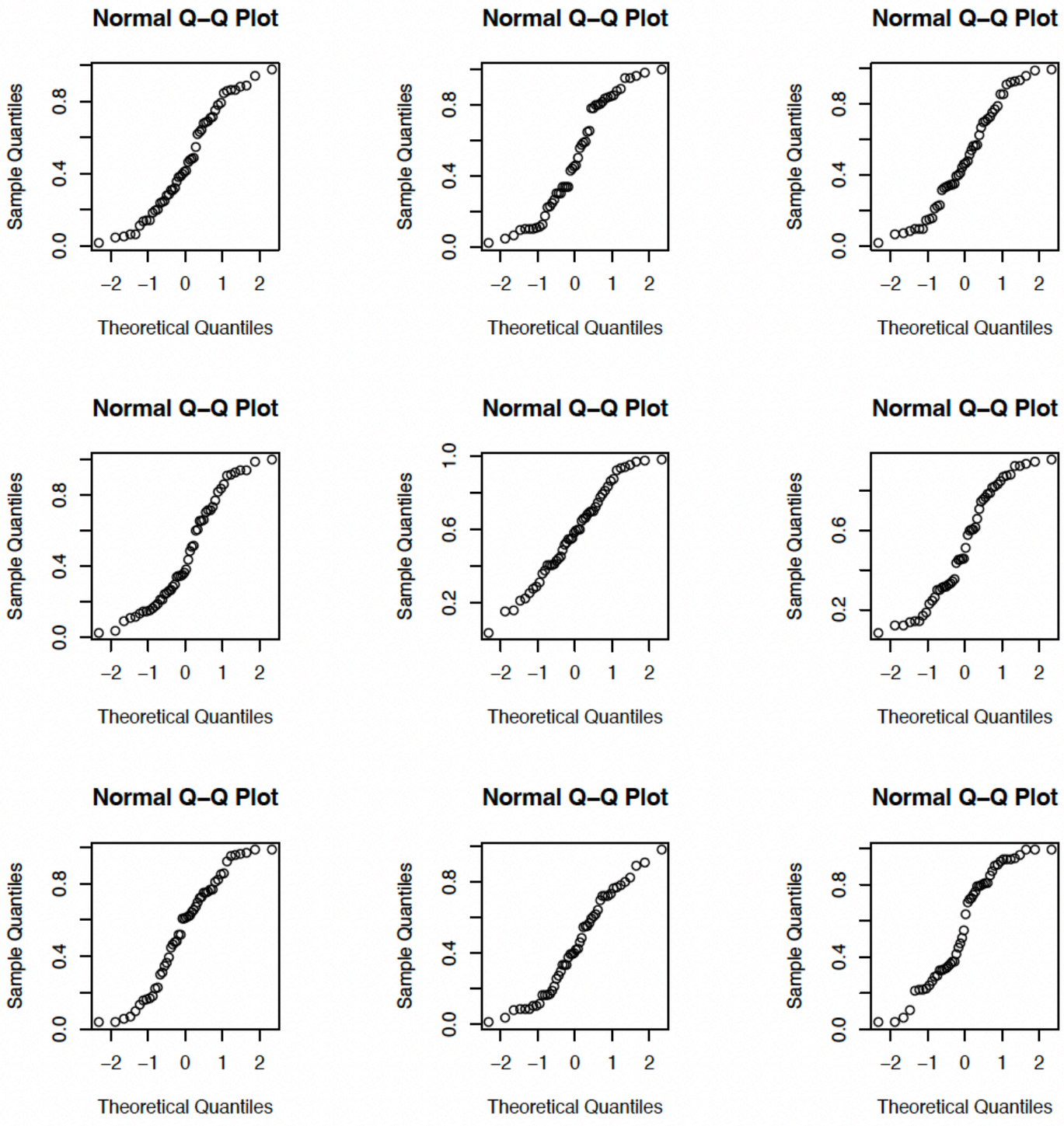

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles

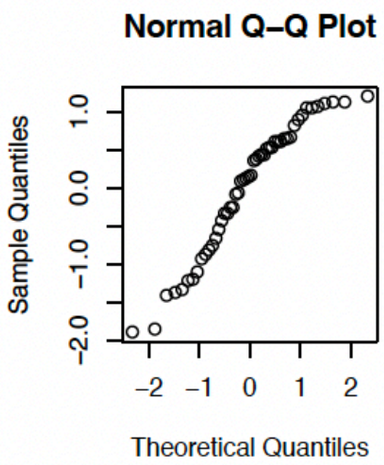
```
## now try Uniform(0,1)

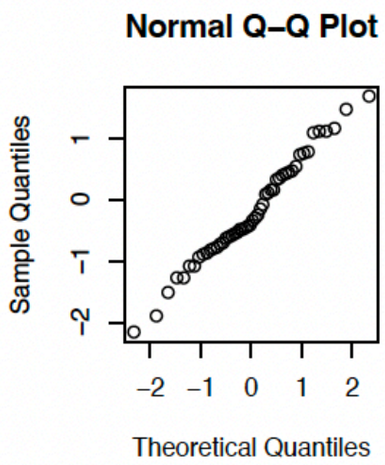
> for(i in 1:9){
>   qqnorm(runif(50))
> }
```

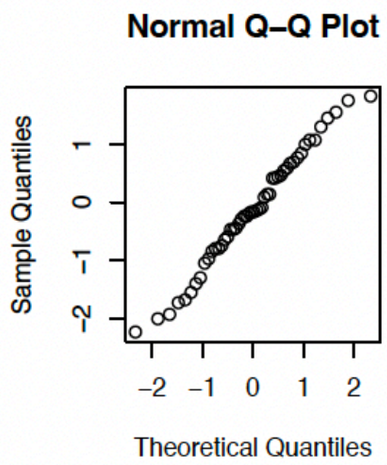

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2 −1 0 1 2
0.0 0.4 0.8
0.0 0.2 0.4 0.6 0.8 1.0

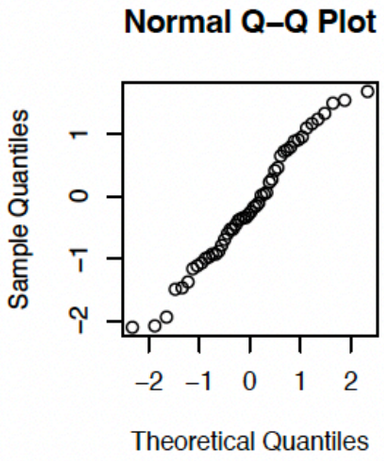
```
## finally try N(0,1)

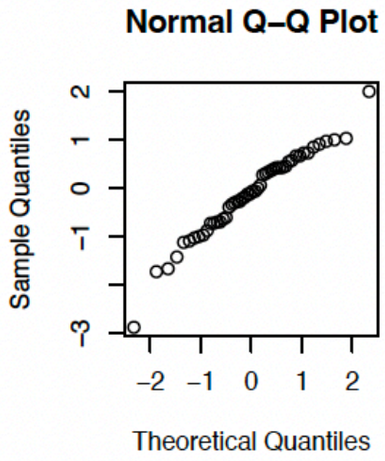
> for(i in 1:9){
>    qqnorm(rnorm(50))
> }
```

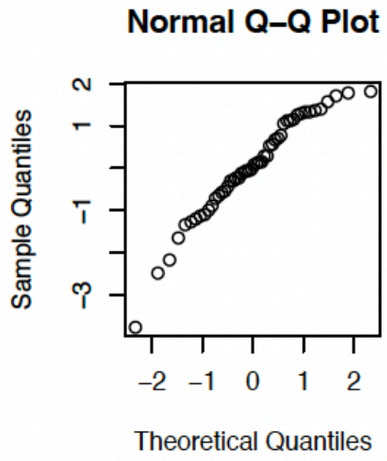

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2.0
−1.0
0.0
1.0
−2
−1
0
1
2

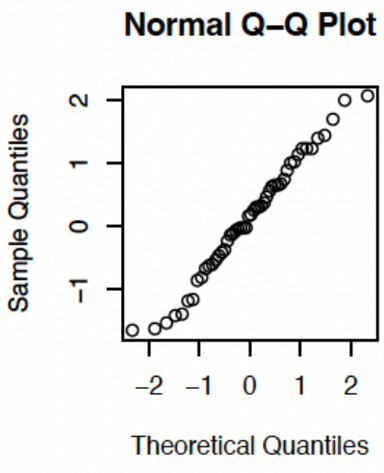

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2

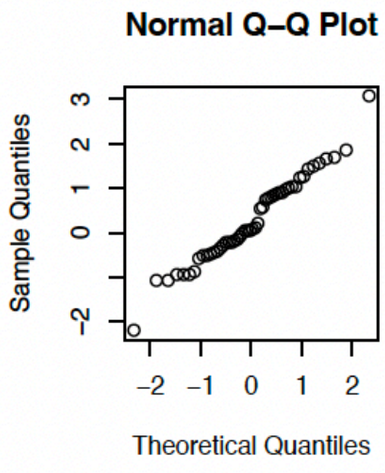

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2

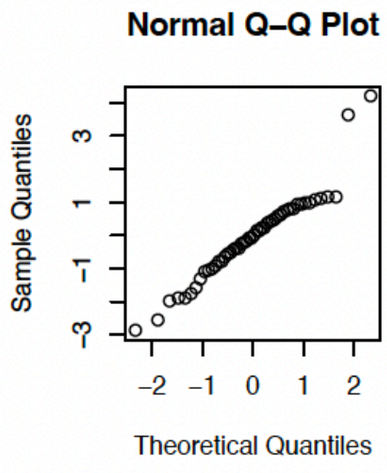

Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−1
1
2
−2
−1
0
1
2


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
−2
−1
0
1
2


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
3
−2
−1
0
1
2


Normal Q–Q Plot
Sample Quantiles
Theoretical Quantiles
−3
−1
1
3
−2
−1
0
1
2

Summary of Diagnostics

- Just fitting a model is not enough
- Graphical diagnostics are more informative but also more subjective
- Diagnostics often suggest a change in the model and then the whole process is repeated
- Time-consuming... but worth it

Thank you!

We gratefully acknowledge the contributions of previous course instructors and various online sources for some of the images and slides presented.